Approximating Self-attention

Target: deriving new RNN states by approximating softmax attention

🍉 Key tools

1. First-order Taylor Expansion $f(x') = f(x+\Delta x) = f(x) + \nabla_x f(x)|_x \cdot \Delta x$
2. Approximation of exponentials
   a. $e^x \approx 1 + x$ near 0
   b. The performer way: $e^{k^T q} = \mathbf{E}_{\omega \in \mathcal{N}(0, \mathbf{I}_d)} e^{k^T w - \|k\|_2/2} \cdot e^{q^T w - \|q\|_2/2}$

🍉 Observations

1. Approximating the exponential => building RNN states.
2. ~~Taylor expansion (of normalizers) => delta rules.~~

Denote the self-attention output (or correspondingly, readout vector of RNN),

$$o_q^i = \sum_{j=1}^{i} \frac{\exp\{k_j^T q\} v_j}{\sum_{l=1}^{i} \exp\{k_l^T q\}} = \sum_{j=1}^{i} \frac{\exp\{k_j^T q\}}{Z_q^i} v_j = \sum_{j=1}^{i} P_q(j) v_j = \mathbf{E}_{P_q}[v] \tag{1}$$

Now, let's consider changes from step i to i+1. We always write $q \triangleq q_i, \quad q' \triangleq q_{i+1}, \quad \Delta q \triangleq q' - q$
,

$$o_{q'}^{i+1} = \sum_{j=1}^{i+1} \frac{\exp\{k_j^T q'\}}{Z_{q'}^{i+1}} v_j = \sum_{j=1}^{i} \frac{\exp\{k_j^T q'\}}{Z_{q'}^{i+1}} v_j + \frac{\exp\{k_{i+1}^T q'\}}{Z_{q'}^{i+1}} v_{i+1} \quad (2)$$

$$= \sum_{j=1}^{i} \underbrace{\frac{\exp\{k_j^T q'\} v_j}{Z_{q'}^{i}} \frac{Z_{q'}^{i}}{Z_{q'}^{i+1}}}_{\text{online softmax trick}} + \frac{\exp\{k_{i+1}^T q'\}}{Z_{q'}^{i+1}} v_{i+1} \quad (3)$$

$$= \underbrace{o_{q'}^{i}}_{\text{part A}} \underbrace{\frac{Z_{q'}^{i}}{Z_{q'}^{i+1}}}_{\text{part B}} + \underbrace{\frac{\exp\{k_{i+1}^T q'\}}{Z_{q'}^{i+1}} v_{i+1}}_{\text{part C}} \quad (4)$$

Part A

For part A, we apply the first-order Taylor expansion,

$$o_{q'}^{i} = o_q^i + \nabla_q o_q^i \cdot \Delta q \quad (5)$$

where we compute the gradient with "log trick" as policy gradients,

$$\nabla_q o_q^i = \nabla_q \mathbf{E}_{P_q}[v] = \nabla_q \sum_{j=1}^{i} P_q(j) v_j = \sum_{j=1}^{i} \nabla_q P_q(j) v_j = \sum_{j=1}^{i} P_q(j) v_j \nabla_q^T \log P_q(j) = \mathbf{E}_{P_q}[v \nabla_q^T \log P_q]($$

$$\nabla_q \log P_q(j) = \nabla_q \log \frac{\exp\{k_j^T q\}}{Z_q^i} = k_j - \nabla_q \log Z_q^i = k_j - \frac{1}{Z_q^i} \nabla_q \sum_{l=1}^{i} \exp\{k_j^T q\} \quad (7)$$

$$= k_j - \sum_{l=1}^{i} \frac{\exp\{k_j^T q\}}{Z_q^i} k_j = k_j - \mathbf{E}_{P_q}[k] \quad (8)$$

Therefore,

$$\nabla_q o_q^i = \sum_{j=1}^{i} P_q(j) v_j (k_j - \mathbf{E}_{P_q}[k])^T = \mathbf{E}_{P_q}[v k^T] - \mathbf{E}_{P_q}[v] \mathbf{E}_{P_q}[k^T] = \mathbf{E}_{P_q}[v k^T] - o_q^i \mathbf{E}_{P_q}[k^T] \quad (9)$$

$$o_{q'}^i = o_q^i + \nabla_q o_q^i \cdot \Delta q + o(\|\Delta q\|) \quad (10)$$

$$\approx o_q^i + \mathbf{E}_{P_q}[v k^T] \Delta q - o_q^i \mathbf{E}_{P_q}[k^T] \Delta q \quad (11)$$

Since the expectation $\mathbf{E}_{P_q}$ still needs to run over sequences, we can further approximate P_q with a uniform distribution $P_q(j) = 1/i$,

> - The approximation here may be improved.

- In policy gradient, the expectation is usually approximated by sampling, that is, $\mathbf{E}_{P_q} v k^T = \frac{1}{m}\sum_{s=1}^{m} v_s k_s^T$
- We may take the policy gradient view and go for the test-time training formulation. That is, as

$$\mathbf{E}_{P_q} v k^T \approx \frac{1}{i}\sum_{j=1}^{i} v_j k_j^T \triangleq S_i, \quad \mathbf{E}_{P_q} k^T \approx \frac{1}{i}\sum_{j=1}^{i} k_j^T \triangleq K_i^T \tag{12}$$

We have,

$$\begin{aligned} o_{q'}^i &\approx o_q^i + \mathbf{E}_{P_q}[v k^T]\Delta q - o_q^i \mathbf{E}_{P_q}[k^T]\Delta q && (13) \\ &\approx o_q^i + S_i \Delta q - o_q^i K_i^T \Delta q && (14) \\ &= o_q^i + \underbrace{(S_i - o_q^i K_i^T)}_{\text{delta rule (not exactly!)}} \Delta q && (15) \end{aligned}$$

Remarks on alternative approximations (those applied in linear attention)

1. ~~We derive a delta rule by taking normalizor~~ Z_q^i ~~in the Taylor expansion~~,

Vanilla Linear Attention:

$$\text{Parallel training:} \quad \boldsymbol{O} = \cancel{\text{softmax}}\left(\boldsymbol{Q}\boldsymbol{K}^\top + \log \boldsymbol{M}\right)\boldsymbol{V} = \left(\boldsymbol{Q}\boldsymbol{K}^\top \odot \boldsymbol{M}\right)\boldsymbol{V} \quad \in \mathbb{R}^{n\times}$$

$$\text{Iterative inference:} \quad \boldsymbol{o}_t = \sum_{j=1}^{t} \frac{\cancel{\exp}(\boldsymbol{q}_t^\top \boldsymbol{k}_j)}{\cancel{\sum_{l=1}^{t}\exp(\boldsymbol{q}_t^\top \boldsymbol{k}_l)}}\boldsymbol{v}_j = \sum_{j=1}^{t}(\boldsymbol{q}_t^\top \boldsymbol{k}_j)\,\boldsymbol{v}_j \quad \in \mathbb{R}^d$$

$\boldsymbol{M}$ 改为线性注意力的causal mask： $\boldsymbol{M}_{i,j} = \begin{cases} 0 & \text{if } j > i \\ 1 & \text{if } j \le i \end{cases}$

RNN-like inference:

$$\begin{aligned} \boldsymbol{o}_t &= \sum_{j=1}^{t} \boldsymbol{v}_j \left(\boldsymbol{k}_j^\top \boldsymbol{q}_t\right) && \boldsymbol{k}_j^\top \boldsymbol{q}_t = \boldsymbol{q}_t^\top \boldsymbol{k}_j \in \mathbb{R} \\ &= \left(\sum_{j=1}^{t} \boldsymbol{v}_j \boldsymbol{k}_j^\top\right)\boldsymbol{q}_t && \text{By associativity} \end{aligned}$$

定义状态 $\boldsymbol{S}_t$，重写线性注意力形式：

$$\boldsymbol{S}_t = \sum_{j=1}^{t} \boldsymbol{v}_j \boldsymbol{k}_j^\top = \boldsymbol{S}_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^\top \in \mathbb{R}^{d\times d}, \quad \boldsymbol{o}_t = \boldsymbol{S}_t \boldsymbol{q}_t \in \mathbb{R}^d$$

2. By performing Tayler expansion at q , instead at 0 , we derive a new state (see discussions on part B/C).

Part B/C

In parts B and C, we also approximate $Z_{q'}^i$ through a first-order Taylor expansion

$$
\begin{aligned}
Z_{q'}^i &= \sum_{j=1}^{i} \exp\{k_j^T q'\} = \sum_{j=1}^{i} \exp\{k_j^T q\} + \nabla_q \sum_{j=1}^{i} \exp\{k_j^T q\} \cdot \Delta q + o(\|\Delta q\|) && (16) \\
&\approx Z_q^i + \sum_{j=1}^{i} \exp\{k_j^T q\} k_j^T \Delta q && (17) \\
&\approx Z_q^i + \sum_{j=1}^{i} (k_j^T q + 1) k_j^T \Delta q \quad \text{// assume } k_j^T q \text{ nears 0, may not be accurate in the sense of approxiation} && (18) \\
&\approx Z_q^i + q^T \left(\sum_{j=1}^{i} k_j k_j^T \right) \Delta q + \left(\sum_{j=1}^{i} k_j^T \right) \Delta q && (19) \\
&= Z_q^i + q^T X_i \Delta q + K_i^T \Delta q && (20)
\end{aligned}
$$

Where $X_i \triangleq \sum_{j=1}^{i} k_j k_j^T$, and we have

$$
\text{part B: } \frac{Z_{q'}^i}{Z_{q'}^{i+1}} = \frac{Z_{q'}^i}{Z_{q'}^i + \exp\{k_i^T q'\}} \tag{21}
$$

$$
\begin{aligned}
& && (22) \\
\text{update states: } Z_{q'}^{i+1} &= Z_{q'}^i + \exp\{k_i^T q'\} && (23) \\
&= Z_q^i + q^T X_i \Delta q + K_i^T \Delta q + \exp\{k_{i+1}^T q'\} && (24) \\
X_{i+1} &= X_i + k_{i+1} k_{i+1}^T && (25)
\end{aligned}
$$

Part C can be obtained from $Z_{q'}^{i+1}$

$$
\text{part C: } \frac{\exp\{k_{i+1}^T q'\}}{Z_{q'}^{i+1}} v_{i+1} \tag{26}
$$

> Alternatively, we can have a simpler expansion at 0 ,
>
> $$
> Z_{q'}^{i+1} = \sum_{j=1}^{i+1} \exp\{k_j^T q'\} \approx \sum_{j=1}^{i+1} (k_j^T q' + 1) = K_{i+1}^T q' + (i+1) \tag{27}
> $$
>
> where we only track K_{i+1} in the update, leaving the second-order states X_i untouched.

Full Picture

🍉 Update Rules

$$
\begin{aligned}
\Delta q &= q_{i+1} - q_i = q' - q && (28) \\
o_{q'}^{i+1} &= \underbrace{o_{q'}^{i}}_{\text{part A}} \underbrace{\frac{Z_{q'}^{i}}{Z_{q'}^{i+1}}}_{\text{part B}} + \underbrace{\frac{\exp\{k_i^T q'\}}{Z_{q'}^{i+1}} v_i}_{\text{part C}} && (29) \\
&= \left(o_q^i + (S_i - o_q^i K_i^T)\Delta q\right) \times \left(1 + \frac{\exp\{k_{i+1}^T q'\}}{Z_q^i + q^T X_i \Delta q + K_i^T \Delta q}\right)^{-1} + \frac{\exp\{k_{i+1}^T q'\}}{Z_q^i + q^T X_i \Delta q + K_i^T \Delta q} v_{i+1} && (30) \\
Z_{q'}^{i+1} &= Z_q^i + q^T X_i \Delta q + K_i^T \Delta q + \exp\{k_{i+1}^T q'\} && (31) \\
S_{i+1} &= \frac{i}{i+1} S_i + \frac{1}{i+1} v_{i+1} k_{i+1}^T && (32) \\
K_{i+1} &= \frac{i}{i+1} K_i + \frac{1}{i+1} k_{i+1} && (33) \\
X_{i+1} &= X_i + k_{i+1} k_{i+1}^T && (34)
\end{aligned}
$$

> Linear attentions: the readout function is multiplicative ($o = f(S, q) = Sq$)
>
> Here, the readout is additive $o = f(S, q, v) = Sq + v$

🍉 Refinement 1

$$
\begin{aligned}
\beta_{i+1} &= \frac{\exp\{k_{i+1}^T q'\}}{Z_q^i + q^T X_i \Delta q + K_i^T \Delta q} && (35) \\
o_{q'}^{i+1} &= (1 - \beta_{i+1}) \left(o_q^i + (S_i - o_q^i K_i^T)\Delta q\right) + \beta_{i+1} v_{i+1} \quad \color{red}{// \text{ if } \beta_{i+1} \text{ is small}} && (36) \\
Z_{q'}^{i+1} &= \exp\{k_{i+1}^T q'\} / \beta_{i+1} + \exp\{k_{i+1}^T q'\} && (37)
\end{aligned}
$$

Or by the simpler approximation,

$$
\begin{aligned}
\Delta q &= q_{i+1} - q_i = q' - q && (38) \\
\beta_{i+1} &= \frac{\exp\{k_{i+1}^T q'\}}{i + K_i^T q'} && (39) \\
o_{q'}^{i+1} &= (1 - \beta_{i+1}) \left(o_q^i + (S_i - o_q^i K_i^T)\Delta q\right) + \beta_{i+1} v_{i+1} \quad \color{red}{// \text{ if } \beta_{i+1} \text{ is small}} && (40) \\
S_{i+1} &= \frac{i}{i+1} S_i + \frac{1}{i+1} v_{i+1} k_{i+1}^T && (41) \\
K_{i+1} &= \frac{i}{i+1} K_i + \frac{1}{i+1} k_{i+1} && (42)
\end{aligned}
$$

Delta Rules

Links to linear attention

We first clarify links to linear attention. We write updates of vanilla linear attention,

$$S_{i+1} = S_i + v_{i+1} k_{i+1}^T \tag{43}$$

Combining with its readout function $o^i = S_i q_i$, we have

$$\begin{aligned} o^{i+1} &= S_{i+1} q_{i+1} = \left(S_i + v_{i+1} k_{i+1}^T\right) q_{i+1} = S_i q_i + S_i \Delta q + v_{i+1} k_{i+1}^T q_{i+1} && (44) \\ &= o^i + S_i \Delta q + (k_{i+1}^T q_{i+1}) v_{i+1} && (45) \end{aligned}$$

On the other hand, our previous result of softmax approximation is,

$$o_{q'}^{i+1} = (1 - \beta_{i+1}) \left(o_q^i + (S_i - o_q^i K_i^T) \Delta q\right) + \beta_{i+1} v_{i+1} \tag{46}$$

Recall that the term $o_q^i + (S_i - o_q^i K_i^T) \Delta q$ is an approximation of the covariance matrix under probability P_q ,

$$\begin{aligned} o_q^i + (S_i - o_q^i K_i^T) \Delta q &\approx o_q^i + \left(\mathbf{E}_{P_q}[v k^T] - \mathbf{E}_{P_q}[v] \mathbf{E}_{P_q}[k^T]\right) \Delta q && (47) \\ &\triangleq o_q^i + C_i \Delta q && (48) \end{aligned}$$

Now we can clarify the link between them

$$\begin{aligned} o^{i+1} &= \left(o^i + S_i \Delta q\right) && + (k_{i+1}^T q_{i+1}) v_{i+1} \quad // \text{ linear attention} && (49) \\ o^{i+1} &= \left(o^i + C_i \Delta q\right) (1 - \beta_{i+1}) && + \beta_{i+1} v_{i+1} \quad // \text{ softmax approximation} && (50) \end{aligned}$$

We see that the above first-order approximation of softmax actually replaces the linear attention state S with a covariance-form state C_i .

- We still don't have the delta rule.
- Roughly, the new states maintain additional "baselines" [1,2]

$$C_i = \underbrace{\mathbf{E}_{P_q}[v k^T]}_{\approx S_i} - \underbrace{\mathbf{E}_{P_q}[v] \mathbf{E}_{P_q}[k^T]}_{\text{baseline}} \tag{51}$$

Question: compared with only updating S , what is the benefit (e.g., numeric/training stability)

Let's try to drop the physical meaning of C_i, and only consider the form of recursion

$$o^{i+1} = \left(o^i + A_i \Delta q\right)\left(1 - \beta_{i+1}\right) + \beta_{i+1} v_{i+1} \quad // \ A_i \text{ doesn't have the meaning of } C_i \tag{52}$$

or, generally,

$$o^{i+1} = \left(o^i + A_i \Delta q\right) \alpha_{i+1} + \gamma_{i+1} v_{i+1} \tag{53}$$

We try to write the readout function of that approximation,

$$\begin{aligned} A_{i+1} &= A_i + \Delta A \qquad (54) \\ o^{i+1} &= A_{i+1} q_{i+1} \qquad (55) \end{aligned}$$

Specifically,

$$\begin{aligned} A_{i+1} q_{i+1} \triangleq o^{i+1} &= \left(o^i + A_i \Delta q\right) \alpha_{i+1} + \gamma_{i+1} v_{i+1} \qquad (56) \\ &= A_i q_{i+1} \alpha_{i+1} + \frac{\gamma_{i+1}}{k_{i+1}^T q_{i+1}} k_{i+1}^T q_{i+1} v_{i+1} \qquad (57) \\ &= \left(A_i \alpha_{i+1} + \frac{\gamma_{i+1}}{k_{i+1}^T q_{i+1}} v_{i+1} k_{i+1}^T\right) q_{i+1} \qquad (58) \end{aligned}$$

We have a gated update of states,

$$A_{i+1} = A_i \alpha_{i+1} + \frac{\gamma_{i+1}}{k_{i+1}^T q_{i+1}} v_{i+1} k_{i+1}^T \tag{59}$$

That is,

🍉 We can have two constraints on the recursive equation

$$o^{i+1} = \left(o^i + A_i \Delta q\right)\left(1 - \beta_{i+1}\right) + \beta_{i+1} v_{i+1} \tag{60}$$

- We endow A_i a physical meaning $A_i = C_i$ is the covariance matrix
- We require an explicit readout function $o^i = A_i q_i$

Question 1: Can we unify the two constraints? That is

- Can we form an estimation of C_i in the form of the readout function requires
- Or can we write a new readout function that satisfies the estimation formula of C_i ?

Question 2: If we can not, what are their intrinsic differences?

Full updates of the covariance matrix

Departing from the previous argument, we can perform the "uniform distribution" approximation directly on the covariance matrix.

$$
\begin{aligned}
C_i &= \mathbf{E}_{P_q}[vk^T] - \mathbf{E}_{P_q}[v]\mathbf{E}_{P_q}[k^T] = \mathbf{E}_{P_q}[(v - \mathbf{E}_{P_q}v)(k - \mathbf{E}_{P_q}k)^T] && (61)\\
D_i &\triangleq \sum_{j=1}^{i}(v_j - \nu_i)(k_j - \kappa_i)^T \quad // \text{ the scatter matrix} && (62)\\
\nu_i &\triangleq \frac{1}{i}\sum_{j=1}^{i} v_j, \quad \kappa_i \triangleq \frac{1}{i}\sum_{j=1}^{i} k_j \quad // \text{ approximate estimation of expectations} && (63)\\
C_i &\approx \frac{1}{i-1}D_i && (64)
\end{aligned}
$$

D_i is the unnormalized covariance estimation.

Now we derive the delta rule by performing an online rank-1 update of D_i (besides online softmax, we also have online covariance estimation!). The algorithm is called Welford's online algorithm. The main job is to write an online update from D_i to D_{i+1} ,

$$
\begin{aligned}
D_{i+1} &= \sum_{j=1}^{i+1}(v_j - \nu_{i+1})(k_j - \kappa_{i+1})^T && (65)\\
&= \sum_{j=1}^{i+1}\left((v_j - \nu_i) - (\nu_{i+1} - \nu_i)\right)\left((k_j - \kappa_i) - (\kappa_{i+1} - \kappa_i)\right)^T && (66)\\
&= D_i - \underbrace{\left(\sum_{j=1}^{i}(v_j - \nu_i)\right)}_{=0}(\kappa_{i+1} - \kappa_i)^T - (\nu_{i+1} - \nu_i)\underbrace{\left(\sum_{j=1}^{i}(k_j - \kappa_i)\right)}_{=0} && (67)\\
&\quad + i(\nu_{i+1} - \nu_i)(\kappa_{i+1} - \kappa_i)^T && (68)\\
&\quad + \underbrace{\left((v_{i+1} - \nu_i) - (\nu_{i+1} - \nu_i)\right)\left((k_{i+1} - \kappa_i) - (\kappa_{i+1} - \kappa_i)\right)^T}_{\text{the incremental term from the new data point i+1}} && (69)
\end{aligned}
$$

From the update of expectation $\nu_i \to \nu_{i+1}$, we have

$$
(i+1)\nu_{i+1} = i\nu_i + v_{i+1} \Rightarrow \nu_{i+1} - \nu_i = \frac{1}{i+1}(v_{i+1} - \nu_i) \qquad (70)
$$

Therefore, the last two terms become

$$i\left(\frac{1}{i+1}\right)^2(v_{i+1}-\nu_i)(k_{i+1}-\kappa_i)^T+\left(\frac{i}{i+1}\right)^2(v_{i+1}-\nu_i)(k_{i+1}-\kappa_i)^T \tag{71}$$

$$=\frac{i}{i+1}(v_{i+1}-\nu_i)(k_{i+1}-\kappa_i)^T \tag{72}$$

We have the delta rule,

$$D_{i+1}=D_i+\frac{i}{i+1}(v_{i+1}-\nu_i)(k_{i+1}-\kappa_i)^T \tag{73}$$

$$=\underbrace{D_i+\frac{i}{i+1}(\nu_i\kappa_i^T-v_{i+1}\kappa_i^T-\nu_i k_{i+1}^T)}_{\text{delta rule}}+\frac{i}{i+1}v_{i+1}k_{i+1}^T \tag{74}$$

Put them all,

$$\begin{aligned}
o_{q'}^{i+1}&=(1-\beta_{i+1})\left(o_q^i+(S_i-o_q^iK_i^T)\Delta q\right)+\beta_{i+1}v_{i+1}\\
&\Downarrow\\
o^{i+1}&=\left(o^i+C_i\Delta q\right)(1-\beta_{i+1})+\beta_{i+1}v_{i+1}\\
&\approx\left(o^i+i^{-1}D_{i+1}\Delta q\right)(1-\beta_{i+1})+\beta_{i+1}v_{i+1}\quad //\text{ if one uses } D_i\text{, no delta rule}\\
&=\left(o^i+i^{-1}\left(D_i+\frac{i}{i+1}(\nu_i\kappa_i^T-v_{i+1}\kappa_i^T-\nu_ik_{i+1}^T+v_{i+1}k_{i+1}^T)\right)\Delta q\right)(1-\beta_{i+1})+\beta_{i+1}v_{i+1}
\end{aligned}$$

Links to deltanet

Now we have,

$$o^{i+1}=\left(o^i+S_i\Delta q\right)\qquad+(k_{i+1}^Tq_{i+1})v_{i+1}\quad //\text{ linear attention} \tag{80}$$

$$o^{i+1}=\left(o^i+S_i\Delta q-S_ik_{i+1}k_{i+1}^Tq_{i+1}\right)\qquad+(k_{i+1}^Tq_{i+1})v_{i+1}\quad //\text{ delta net} \tag{81}$$

$$\tag{82}$$

$$o^{i+1}=\left(o^i+C_i\Delta q\right)(1-\beta_{i+1})\qquad+\beta_{i+1}v_{i+1}\quad //\text{ softmax approx. w. covariance} \tag{83}$$

$$o^{i+1}=\left(o^i+(i-1)^{-1}D_i\Delta q\right)(1-\beta_{i+1})\qquad+\beta_{i+1}v_{i+1}\quad //\text{ softmax approx. w.o. delta rule} \tag{84}$$

$$o^{i+1}=\left(o^i+i^{-1}D_{i+1}\Delta q\right)(1-\beta_{i+1})\qquad+\beta_{i+1}v_{i+1}\quad //\text{ softmax approx. w. delta rule} \tag{85}$$

Recall that we can write the readout function

$$A_{i+1} = A_i \alpha_{i+1} + \frac{\gamma_{i+1}}{k_{i+1}^T q_{i+1}} v_{i+1} k_{i+1}^T \tag{86}$$

$$\tag{87}$$

$$D_{i+1} = D_i + \frac{i}{i+1}(v_{i+1} - \nu_i)(k_{i+1} - \kappa_i)^T \tag{88}$$

$$= D_i + \frac{i}{i+1}(\nu_i \kappa_i^T - v_{i+1}\kappa_i^T - \nu_i k_{i+1}^T) + \frac{i}{i+1} v_{i+1} k_{i+1}^T \tag{89}$$

$$\Rightarrow \tag{90}$$

$$C_{i+1} = \frac{1}{i} D_{i+1} \tag{91}$$

$$= (1 - \frac{1}{i})C_i + \frac{1}{i+1}(\nu_i \kappa_i^T - v_{i+1}\kappa_i^T - \nu_i k_{i+1}^T) + \frac{1}{i+1} v_{i+1} k_{i+1}^T \tag{92}$$

Full picture with the delta rule

Upon substitution, we obtain

🍉 State

$$\nu_t = \frac{1}{t}\sum_{j=1}^{t} v_j \tag{93}$$

$$\kappa_t = \frac{1}{t}\sum_{j=1}^{t} k_j \tag{94}$$

$$D_t \triangleq \sum_{j=1}^{t}(v_j - \nu_t)(k_j - \kappa_t)^\top \tag{95}$$

Recurrence

$$\nu_t = \nu_{t-1} + \frac{1}{t}(v_t - \nu_{t-1}) \tag{96}$$

$$\kappa_t = \kappa_{t-1} + \frac{1}{t}(k_t - \kappa_{t-1}) \tag{97}$$

$$D_t = D_{t-1} + \frac{t-1}{t}(v_t - \nu_{t-1})(k_t - \kappa_{t-1})^\top \tag{98}$$

Sequence Parallel

$$P_t = \mathrm{PrefixSum}(v)_t \tag{99}$$
$$R_t = \mathrm{PrefixSum}(k)_t \tag{100}$$
$$M_t = \mathrm{PrefixSum}(vk^\top)_t \tag{101}$$
$$D_t = M_t - \frac{1}{t} P_t R_t^\top \tag{102}$$

Output

$$\begin{cases} \Delta q_t = q_t - q_{t-1} & (103) \\ \beta_t = \dfrac{\exp(k_t^\top q_t)}{(t-1) + R_{t-1}^\top q_t} & (104) \\ o_t = (1-\beta_t)(o_{t-1} + (i-1)^{-1} D_t \Delta q_t) + \beta_t v_t & (105) \end{cases}$$

Designs

Variational view of softmax

Links to SSM (from the first-order Taylor expansion to the discretization of ODE)